

llmXive Automated Review of Formalizing Latent Thoughts: Four Axioms of Thought Representation in LLMs

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the accuracy of factual claims and their alignment with the provided data tables and citations.

****1. Metric Definition and Data Consistency (Appendix e000, Section 'Geometric Analysis of Representational Collapse'):** The text states: "Soft Thinking with Gumbel noise lands at $\Delta_{\cos} \in [0.02, 0.05]$, indistinguishable from Random Vector." ****Issue:** The metric $\Delta_{\cos}$ is not defined in the surrounding text or the main methodology section. The paper primarily reports DCS (Discriminator-based Causal Stability) as AUROC in Table 'tab:stability-headline'. In that table, Random Vector (RV) scores are consistently around 0.50–0.56 (random chance), while Soft Thinking with Gumbel noise (STN) scores are significantly higher (0.84–0.90). ****Discrepancy:** If $\Delta_{\cos}$ refers to a cosine similarity difference, the claim that STN is "indistinguishable" from RV contradicts the DCS AUROC results where STN is clearly distinct from RV. If $\Delta_{\cos}$ is a new metric not reported in the tables, it must be defined and its values provided. If the authors intended to refer to the DCS metric, the claim is factually incorrect based on the provided tables. ****Action:** Define $\Delta_{\cos}$ or correct the claim to align with the reported DCS AUROC values.

****2. Statistical Methodology (Appendix e000, Section 'Bootstrap Confidence Intervals'):** The text states: "A 95% interval is recovered as $\mu \pm 1.96 \widehat{\sigma}_B$." ****Issue:** The section begins by stating the standard error is obtained by the "percentile method," but the formula provided ($\mu \pm 1.96 \sigma$) describes a

parametric (normal-approximation) interval. These are two different methods. The percentile method uses the 2.5th and 97.5th percentiles of the bootstrap distribution directly, not the mean ± 1.96 standard errors. **Impact:** If the distribution of the bootstrap estimates is skewed (common for accuracy or AUROC near boundaries), the parametric formula yields incorrect intervals. **Action:** Clarify whether the percentile method or the normal approximation was used. If the percentile method was used, the formula description is inaccurate. If the normal approximation was used, the mention of the "percentile method" is misleading.

3. Citation and Claim Alignment (Section 'Results'): The text claims: "No candidate exceeds IE on every axis across LLMs." **Observation:** This claim appears supported by the data in Tables 'tab:causality-results', 'tab:minimality-headline', 'tab:separability-headline', and 'tab:stability-headline'. For example, while Soft Thinking (STN) beats Input Embedding (IE) on Minimality for Llama 8B (0.24 vs 0.22), it loses on Causality (5.08 vs 4.71, where lower is better) and Separability (53.5 vs 54.5). **Recommendation:** While the claim is factually supported by the aggregate data, the text would benefit from explicitly citing the specific table cells or providing a summary sentence that clarifies the trade-offs (e.g., "While STN improves Minimality on Llama 8B, it degrades Causality and Separability compared to IE"). This prevents ambiguity regarding the directionality of the metrics (e.g., whether higher or lower is better for each axis).

4. Citation Validity: The citations (e.g., 'wu2026llms', 'zhang2025softthinkingunlockingreasoning') refer to future-dated papers (2025, 2026). As this is a review of a preprint from a public archive (arXiv 2606.27378), these citations are consistent with the paper's internal timeline (2026). No factual error is detected in the citation keys themselves relative to the provided bibliography, assuming the bibliography is complete. However, the claim that "Soft Thinking... ends up being greedy decoding" (citing 'wu2026llms') should be verified against the actual content of that cited work if available, but based on the provided text, the citation is used consistently to support the claim about noise necessity.

Conc

- [science] Appendix e000 claims Soft Thinking (Gumbel) is 'indistinguishable from Random Vector' with $\Delta_{\cos}$ in $[0.02, 0.05]$. However, Table tab:stability-headline shows STN AUROC ~ 0.85 vs RV ~ 0.50 . Define $\Delta_{\cos}$ or correct the claim to match reported DCS data.
- [writing] Appendix e000 states 95% CI is ' $\mu \pm 1.96 \cdot \sigma_B$ ' but claims to use the 'percentile method'. These are contradictory. Clarify if the normal approximation or percentile method was actually used for the reported intervals.
- [writing] Section 'Results' claims 'No candidate exceeds IE on every axis'. While supported by tables, explicitly cite the specific table cells (e.g., tab:minimality-headline vs tab:causality-results) to clarify metric directionality (higher/lower is better) and avoid ambiguity.

paper_reviewer_figure_critic

Verdict: major_revision_science

Figure 1

Figure 1 presents four CDF panels with clear axes and legends, but the top-left 'Anchors' panel suffers from visual clutter where multiple legend entries (Exact vs. Pooled Output Embedding, and the two Last Input Token variants) map to nearly identical or overlapping curves, reducing interpretability.

Figure 2

The figure effectively displays ICC values across different configurations, but the caption incorrectly describes the x-axis grouping (thinking steps vs. averaging windows), and the y-axis label contains unrendered LaTeX code.

Figure 3

The figure is visually clear but fails to render the 95% confidence interval bands explicitly mentioned in the caption. Additionally, the 'Last Input Token' series appears truncated or missing data points for lengths below 64, which contradicts the caption's description of varying substituted length L .

Figure 4

The figure effectively displays correlation data, but the caption contains a broken cross-reference ('order of .') that leaves the x-axis categories undefined, and the y-axis labels are phrased awkwardly.

Figure 5

The figure effectively visualizes the window sweep across different representation families, but the caption's claim of CI bands is not visually supported in the 'Baselines' panel, and axis labeling is inconsistent between rows.

Figure 6

Figure 6 is a clear, self-contained schematic that effectively visualizes the four axiomatic properties (Causality, Minimality, Separability, Stability) described in the paper. The diagram uses distinct panels and intuitive icons to represent abstract concepts without requiring external data or complex statistical annotations, and the caption accurately reflects the content.

Figure 7

The figure is generally clear and supports the claims, but the caption's reference to the threshold 'tau' is not visually represented in the bottom row text or the legend of the third panel, creating a minor inconsistency.

Figure 8

Figure 8 presents clear data but suffers from a contradiction between the caption's claim of a fixed threshold and the varying thresholds shown in panel titles. Additionally, the bottom row legend is incomplete, missing the 'Soft Thinking - Gumbel' entry, and the top row x-axis labels are too dense to read easily.

Figure 9

The figure effectively demonstrates the stability of rankings across the threshold pa-

parameter, but it suffers from presentation issues including a missing y-axis label, a legend that obscures data in the final panel, and a caption that fails to explicitly define the x-axis symbol $\square$.

Figure 10

The figure effectively visualizes the correlation between discriminator accuracy and task performance, but the caption contains LaTeX formatting errors and the legend does not fully reflect the five LLMs mentioned in the text.

Figure 11

The figure effectively visualizes the geometric properties of thought representations across models, but it suffers from missing axis labels on the top row and right column panels. Additionally, the legend could be clearer in distinguishing between static candidates and the step-count trajectories represented by the lines.

Figure 12

The figure effectively visualizes the output length distributions and supports the caption's claim regarding the order-of-magnitude difference between model families. However, the caption fails to describe the 'mean' symbol explicitly shown in the legend, and the x-axis labels are rotated unnecessarily.

- [writing] Figure 1: The top-left panel ('Anchors') contains a legend with six entries, but the 'Exact Output Embedding' and 'Pooled Output Embedding' lines are visually indistinguishable (overlapping) in the plot, making it impossible to verify if they differ.
- [writing] Figure 1: The top-left panel ('Anchors') legend lists 'Last Input Token (final layer)' and 'Last Input Token (all layers)' as distinct entries, but the corresponding purple lines are nearly identical, raising questions about the visual distinction or the necessity of separate legend entries.
- [science] Figure 2: The legend defines 'W' as the averaging window size, but the x-axis labels (e.g., 'Soft Think. 1', 'Soft Think. 16') represent the number of thinking steps. The caption states the figure shows 'bars per averaging window', yet the plot groups bars by thinking step and colors them by window size, contradicting the caption's description of the grouping.
- [writing] Figure 2: The y-axis label uses LaTeX formatting ($\sigma^2_{\text{between}} / (\sigma^2_{\text{between}} - \sigma^2_{\text{within}})$) that renders as raw code rather than formatted math, making it difficult to read.
- [science] Figure 3: The caption states the plot includes '95% cluster-bootstrap CI bands', but the rendered figure displays only lines and markers with no visible error bands or shaded regions.
- [science] Figure 3: The legend lists 'Last Input Token (all layers)' (pink), but the plot shows this series starting abruptly at $x=64$ with no data points at lower lengths, despite the caption implying a comparison across varying L .
- [writing] Figure 4: The caption states 'Candidates follow the order of .' but the reference

is missing; the x-axis labels (Exc, Pool, All, Fin, 1, 16, etc.) are not defined in the caption or the figure itself.

- [writing] Figure 4: The y-axis label 'r vs. input length' (top) and 'r vs. output length' (bottom) is grammatically incorrect and ambiguous; it should likely be 'Pearson r with input/output length'.
- [writing] Figure 4: The x-axis labels are crowded and difficult to read, particularly in the 'Soft Thinking' and 'Soft T. (Gumbel)' sections where numbers (1, 16, 32, 64, 128) are tightly packed.
- [writing] Figure 5: The caption states the plot shows 'Mean causality KL', but the y-axis is labeled only 'Mean KL', omitting the specific 'causality' descriptor for precision.
- [writing] Figure 5: The x-axis label 'Tail window (tokens)' is present only on the bottom row; the top row panels lack this label, creating visual inconsistency.
- [science] Figure 5: The caption promises '95% bootstrap CI bands', but the 'Baselines' panel (bottom right) displays only single lines without the corresponding shaded error bands.
- [writing] Figure 7: The caption states the bottom row shows distributions 'with the threshold $\tau = 0.9$ overlaid', but the symbol ' τ ' is missing from the text in the image (showing only '= 0.9') and the legend in the rightmost panel is missing the Greek letter ' τ '.
- [writing] Figure 7: The caption mentions 'one panel per source model' for the top row, but the panels are labeled with model names (Llama-3.1 8B, Llama-3.3 70B, DS-R1-Qwen 32B) rather than explicitly stating which 'source model' corresponds to which panel in the text description.
- [writing] Figure 8: The caption states the top row shows AUROC 'at $\tau = 0.9$ ', but the individual panel titles display varying thresholds (e.g., 'Hx > 0: 28.2%', 'Hx > 0: 1.0%'), creating a contradiction between the summary text and the visual data.
- [writing] Figure 8: The top row x-axis labels are rotated 90 degrees and extremely dense, making the method names (e.g., 'Thinking@32 - Gumbel@2') difficult to read without zooming.
- [science] Figure 8: The bottom row legend defines 'Soft Thinking' and 'Latent Thinking' but omits the 'Soft Thinking - Gumbel' family, which is clearly plotted in the Skywork-OR1 and GPT-OSS panels.
- [writing] Figure 9: The caption refers to 'semantic equivalence threshold τ ' but the x-axis is labeled with the symbol ' τ ' (tau); the caption should explicitly define τ as the threshold.
- [writing] Figure 9: The legend is located inside the rightmost panel (GPT-OSS 20B) and obscures the data lines; it should be moved outside the plot area or made semi-transparent.

- [writing] Figure 9: The y-axis label 'AUROC' is missing; while the caption mentions AUROC, the axis itself is unlabeled, making the plot standalone interpretation difficult.
- [writing] Figure 10: The caption contains LaTeX formatting artifacts ('Spearman \$\$') instead of the rendered symbol (ρ) shown in the plot.
- [writing] Figure 10: The legend in the right panel lists 'Llama-3.1 8B' and 'Llama-3.3 70B', but the caption states the data is averaged across five LLMs; the legend does not account for the other three models.
- [science] Figure 11: The legend defines 'Soft Thinking (1-128 steps)' as a single entry, but the blue line in the plots is annotated with '1' at the start and '128' at the end. The caption states lines trace trajectories as step count grows, but the legend fails to explicitly indicate that the lines represent a range of steps rather than a single static comparison, which may confuse readers regarding the data density.
- [writing] Figure 11: The x-axis label 'within-task participation ratio PR' is only present on the bottom row of plots. The top row panels (Llama-3.1-8B-Instruct, Llama-3.3-70B-Instruct, DeepSeek-R1-Distill-Qwen-32B) lack x-axis labels, forcing the reader to infer the axis meaning from the bottom row.
- [writing] Figure 11: The y-axis label 'k-NN task purity' is only present on the bottom-left plot. The other four panels lack y-axis labels, which is poor practice for multi-panel figures even if the axis is shared.
- [writing] Figure 12: The orange diamond symbol is defined in the legend as 'mean', but the caption text only discusses the 'median' (box line) and does not mention the mean, creating a disconnect between the visual data and the textual description.
- [writing] Figure 12: The x-axis labels are rotated at a steep angle, which is unnecessary for the short model names and reduces readability compared to a horizontal layout.

paper_reviewer_jargon_police

Verdict: minor_revision

The manuscript relies heavily on specialized shorthand and undefined acronyms that create a barrier for non-specialist readers. In Section 3 (Table 1) and the Appendix, the term **DCS** is used repeatedly without being spelled out as "Discriminator-based Causal Stability" or similar, forcing the reader to guess its meaning from context. Similarly, **TR** appears in Appendix A.2 ("TR-independent constant") without definition; it should be explicitly stated as "Thought Representation" upon first use.

In Appendix A.3, the statistical metric **ICC** is introduced to interpret Figure 2 without definition. While standard in statistics, it is not universally known in all ML subfields and should be expanded to "Intraclass Correlation Coefficient" with a brief parenthetical explanation of its relevance to the analysis. Additionally, **UF2** in Appendix A.5 ("Deep-UF2") is an opaque acronym for "unfrozen blocks" that should be defined immediately to avoid

confusion.

Finally, the benchmark **BBEH** is referenced frequently (e.g., Section 4) but the full name is not explicitly written out in the main text, relying solely on the citation. Defining these terms at their first occurrence would significantly improve accessibility without sacrificing technical precision.

- [writing] The manuscript relies heavily on specialized shorthand and undefined acronyms that create a barrier for non-specialist readers. In Section 3 (Table 1) and the Appendix, the term DCS is used repeatedly without being spelled out as "Discriminator-based Causal Stability" or similar, forcing the reader to guess its meaning from context. Similarly, TR appears in Appendix A.2 ("TR-independent constant") without definition; it should be explicitly stated as "Thought Representation" upon first use. In A

paper_reviewer_logical_consistency

Verdict: minor_revision

The logical consistency of the paper is compromised by a disconnect between the abstract axiomatic definitions and the specific empirical metrics used to evaluate them.

First, the central conclusion that "No candidate satisfies all four axioms" (Introduction) does not strictly follow from the data. The paper defines "Stability" as the ability to encode entropy and "Separability" as linear discriminability. The results show that Output Embedding (OE) achieves high scores on both (DCS ~ 0.96 , Separability $\sim 72\%$). The paper dismisses OE based on "Minimality," but the Minimality metric (Δ_{IB}) is a relative score compared to baselines, not a binary check of the sufficient statistic property. The authors conflate "no candidate is the global optimum across all metrics" with "no candidate satisfies the logical axioms." If OE satisfies Causality, Separability, and Stability, and is a sufficient statistic (by definition of being the output), it logically satisfies the axioms, even if it is not minimal in the Information Bottleneck sense relative to the input. The conclusion overreaches the evidence.

Second, the proof of "Independence" in the Appendix (app:independence) constructs counter-models that violate the *operationalization* of the metrics rather than the axioms themselves. For instance, the "Violation of Causality" counter-model permutes coordinates of $\mathbf{T}^*$. This breaks the specific KL substitution test (which relies on index alignment) but does not necessarily break the abstract causal link between $\mathbf{T}$ and $\mathbf{Y}$. The paper assumes that if the metric fails, the axiom is violated, but the metric is a specific proxy, not the axiom itself. This creates a circular logic where the axioms are defined by the metrics, and the metrics are used to prove the axioms are independent.

Finally, the interpretation of the "Stability" metric is logically inverted. The axiom requires $\mathbf{T}$ to encode the entropy of the semantic distribution. The metric (DCS) measures the ability to *detect* high entropy (uncertainty). The results show iterative methods (Soft Thinking) have *lower* DCS scores as steps increase. The authors interpret this as a loss of stability. However, logically, if the model is "thinking" and converging on an

answer, the entropy of the semantic distribution should **decrease** (becoming more certain). A lower DCS score in this context would indicate the model is successfully reducing uncertainty, which is a **positive** sign of reasoning, not a violation of stability. The paper's conclusion that iterative methods "degrade" in stability contradicts the logical expectation that reasoning reduces entropy. The metric measures the **presence** of uncertainty, not the **stability** of the representation under perturbation, leading to a misinterpretation of the results.

- [writing] The logical consistency of the paper is compromised by a disconnect between the abstract axiomatic definitions and the specific empirical metrics used to evaluate them. First, the central conclusion that "No candidate satisfies all four axioms" (Introduction) does not strictly follow from the data. The paper defines "Stability" as the ability to encode entropy and "Separability" as linear discriminability. The results show that Output Embedding (OE) achieves high scores on both (DCS ~ 0.96 , Separ

paper_reviewer_overreach

Verdict: minor_revision

The paper proposes a rigorous axiomatic framework for evaluating latent thought representations in LLMs. While the theoretical construction is sound, the manuscript occasionally overreaches in its interpretation of the empirical results and the scope of its theoretical claims.

First, the claim of "Completeness" in Appendix \ref{app:completeness} is presented as a definitive property of the axiom set $\mathcal{P}$. The proof relies on an "idealized model" $\mathcal{M}^*$ and an "idealized limit" where stability holds with equality. However, the main text and results (Section \ref{sec:results}) demonstrate that no existing candidate satisfies these axioms in practice. The paper risks misleading readers by implying the axioms are a complete description of **actual** thought in current LLMs, rather than a complete description of a **theoretical ideal** that current models fail to reach. The distinction between the theoretical completeness of the definition and the empirical incompleteness of the candidates needs sharper delineation to avoid overclaiming the framework's immediate applicability.

Second, the conclusion asserts that the protocol "exposes representational collapse on per-question identity not revealed by downstream accuracy." This phrasing overstates the causal inference. The data in Appendix \ref{appendix:downstream} shows a near-zero correlation ($\rho = 0.10$) between the discriminator's ability to separate tasks and the model's downstream pass@1 accuracy. While this indicates that high accuracy does not guarantee a well-structured latent space, it does not definitively prove that "representational collapse" is the specific mechanism causing the disconnect, nor does it prove that accuracy metrics are entirely blind to the phenomenon. The claim should be tempered to reflect that the metrics reveal a **divergence** between task performance and representational structure, rather than a definitive exposure of a specific failure mode like "collapse"

that accuracy metrics miss.

Finally, the framing of iterative methods (Soft Thinking, Latent Thinking) as candidates for satisfying the axioms is slightly overreaching given the results. The data shows that as step counts increase, these methods often degrade in Stability and Separability (Section \ref{sec:results:per-family}). The paper implies these methods are approaching the axiomatic ideal, but the evidence suggests they may be moving away from it. The discussion should more honestly address whether these methods are fundamentally incompatible with the proposed axioms (specifically Stability) rather than framing them as simply "not yet" satisfying them.

- [writing] The claim that the four axioms are 'complete' (Appendix:app:completeness) is an overreach. The proof asserts a 'functional isomorphism' based on idealized limits, yet the empirical results (Section 5) show no candidate satisfies all axioms simultaneously. The paper must clarify that 'completeness' applies only to the theoretical construct, not the current empirical reality of LLMs.
- [writing] The conclusion states the protocol 'exposes representational collapse on per-question identity not revealed by downstream accuracy.' This overstates the finding. The data (Appendix:appendix:downstream) shows near-zero correlation ($\rho=0.10$) between discriminator accuracy and task pass@1, but this does not prove 'collapse' is the *cause* of the lack of correlation, nor that accuracy metrics are blind to it. The causal language should be softened to 'suggests a disconnect'.
- [writing] The abstract and introduction imply the axioms define a 'Functional Thought Representation' that LLMs *should* possess. However, the results show current methods (Soft Thinking, Latent Thinking) often degrade as steps increase. The paper overreaches by framing these methods as attempts to satisfy the axioms when the data suggests they may be fundamentally misaligned with the 'Stability' and 'Minimality' requirements as defined.

paper_reviewer_safety_ethics

Verdict: minor_revision

The manuscript presents a diagnostic framework for evaluating latent thought representations in LLMs. From a safety and ethics perspective, the primary concern lies in the potential dual-use nature of the proposed "Four Axioms" (Causality, Minimality, Separability, Stability).

In the **"Broader Impacts"** section (Appendix Impact), the authors state: "Diagnostic protocol only; no negative societal effects identified." This assertion is insufficient. The ability to rigorously measure and identify "representational collapse" or specific latent states could be weaponized. For instance, an adversary could use these metrics to optimize a model to produce outputs that appear "stable" and "separable" to the diagnostic probes while actually encoding deceptive or harmful reasoning (a form of "sleeper agent" optimization). The paper should explicitly discuss how these diagnostic tools might be

misused to bypass safety alignment or to engineer models that hide malicious intent within "collapsed" representations.

Regarding **Data and Privacy**, the study utilizes the BBEH benchmark, which includes tasks like "Social Reasoning" and "SARC Triples." While the benchmark is public, the authors should include a brief statement confirming that the evaluation protocol respects the original data licenses and that the beam-search generation process did not inadvertently expose or leak any sensitive or private information that might be present in the training data of the source models or the benchmark itself.

Finally, the **Reproducibility** section (Appendix Reproducibility) directs readers to an external repository. While external hosting is standard, the safety review requires assurance that the released code does not include scripts designed to exploit the identified vulnerabilities (e.g., tools to force "collapse" in a target model). A statement clarifying that the code is strictly for diagnostic measurement and not for adversarial manipulation is recommended.

The paper does not involve human subjects (IRB/IACUC not applicable), but the potential for the methodology to inform adversarial attacks on LLM safety mechanisms necessitates a more robust discussion in the Broader Impacts section.

- [writing] The 'Broader Impacts' section (Appendix Impact) dismissively states 'no negative societal effects identified' without addressing the dual-use risk of the proposed diagnostic protocol. If these axioms can identify 'representational collapse' or specific latent states, malicious actors could use this to engineer models that bypass safety filters by optimizing for 'stable' but deceptive representations. This requires a substantive discussion of potential misuse.
- [writing] The paper evaluates models on sensitive reasoning tasks (e.g., 'Causal Understanding', 'Social Reasoning', 'SARC Triples') using the BBEH benchmark. While the data is public, the manuscript lacks a statement confirming that the evaluation protocol adheres to the original benchmark's data usage licenses and that no private or personally identifiable information was inadvertently exposed during the beam-search generation and analysis.
- [writing] The 'Reproducibility' section links to an external GitHub repository. The safety review cannot verify if the released code contains scripts that could be used to extract or exploit the specific latent representations identified as 'collapsed' or 'unstable'. A statement confirming that the released code is limited to the diagnostic metrics and does not include tools for adversarial manipulation of these states is required.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The paper presents a comprehensive empirical audit of latent thought representations against four axioms, utilizing a large-scale dataset (4,520 problems across 23 tasks) and multiple LLMs. The sample size is robust, and the use of 10,000 bootstrap resamples for

confidence intervals (Appendix e000) is a strong methodological choice. However, the strength of the central claims regarding "representational collapse" and the failure of current methods to satisfy the axioms is somewhat undermined by the presentation of statistical evidence.

First, the claim that no candidate satisfies all axioms hinges on the Input Embedding (IE) baseline failing the Stability (DCS) axiom. Yet, Table 4 (e002) reports DCS AUROC scores for IE ranging from 0.92 to 0.94 across most models, which are numerically high. The text asserts these represent a failure, but without a clear threshold or a comparison to a theoretical maximum, the interpretation of "collapse" is ambiguous. If 0.92 is considered a failure, the definition of the Stability metric's success criteria needs explicit justification in the main text.

Second, the Causality results (Table 2, e001) show very small effect sizes between candidates (e.g., 4.56 vs 4.71 nats for Llama 70B). While the authors use bootstrap standard errors, the main tables omit these error bars. Without seeing the confidence intervals or the results of paired statistical tests (e.g., paired t-tests or Wilcoxon signed-rank tests on the per-problem KL values), it is difficult to determine if the observed differences are statistically significant or merely noise. The claim that "No candidate consistently exceeds Input Embedding" is strong; the evidence provided in the tables is currently insufficient to rule out random variation given the small margins.

Third, the Separability analysis (Table 4) reports "Same-task" accuracy near random chance (50-54%) for most iterative methods. The authors interpret this as structural collapse. However, the Output Embedding (OE) achieves ~63-72% in the same setting. While this suggests OE is better, the gap is not massive, and the lack of a "perfect" representation baseline makes it hard to quantify the severity of the collapse. The paper would benefit from a clearer statistical comparison of the "Same-task" accuracy distributions between the best candidate (OE) and the iterative methods to confirm the collapse is not just a marginal difference.

Finally, the geometric analysis (Appendix e000) mentions that Soft Thinking with Gumbel noise is "indistinguishable from Random Vector" based on $\Delta_{\cos} \in [0.02, 0.05]$. This is a critical finding for the "Stability" and "Separability" axioms. However, the main text does not explicitly link this geometric indistinguishability to the failure of the axioms in a way that is immediately clear to a reader skimming the results. The connection between the geometric metrics (PR, purity) and the axiomatic scores needs to be more tightly integrated in the results section to support the "collapse" narrative.

In summary, the experimental design is sound, but the statistical rigor in the presentation of the results (specifically regarding effect sizes, significance testing, and baseline definitions) needs strengthening to fully support the strong claims of representational collapse.

- [science] The claim that 'No candidate satisfies all four axioms' relies on the Input Embedding (IE) baseline failing Stability (DCS). However, Table 4 shows IE DCS scores of 0.92-0.94 for most models, which are high. The text must explicitly clarify why these high scores are considered a failure of the Stability axiom or if the 'collapse' refers to a

different metric not clearly distinguished in the summary.

- [science] The Causality metric (KL divergence) is reported as a single mean per cell in Table 2, but the text claims 'No candidate consistently exceeds Input Embedding.' Given the small effect sizes (e.g., 4.56 vs 4.71 in Llama 70B) and the lack of per-problem paired statistical tests (e.g., Wilcoxon signed-rank) in the main text, the robustness of this ranking is unclear.
- [science] The Separability results (Table 4) show 'Same-task' accuracy near random (50-54%) for most candidates. The paper attributes this to 'representational collapse.' However, without a control showing that a known good representation (e.g., the Output Embedding) achieves significantly higher same-task accuracy in the *same* experimental setup, the baseline for 'good' separability is ambiguous.
- [science] The bootstrap confidence intervals ($B=10,000$) are mentioned in the appendix, but the main tables (e.g., Table 2, 4) do not display these intervals or standard errors for the headline metrics. To verify the 'no consistent advantage' claim, the reader needs to see the error bars to assess if the differences between candidates are statistically significant or within noise.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical analysis framework is generally robust, utilizing a large number of bootstrap resamples ($B=10,000$) to estimate uncertainty, which is appropriate for the complex metrics involved (KL divergence, Information Bottleneck residuals). The explicit definition of the statistical unit as the "problem" and the use of cluster bootstrapping for the Causality metric (Section 'appendix:sub:bootstrap') are strong practices that account for the hierarchical nature of the data (beams within problems).

However, several statistical reporting details require clarification to ensure the validity of the inferences. First, the handling of the "length gate" in the Causality analysis (Section 'appendix:sub:bootstrap') reduces the effective sample size to 360 problems for Llama-3.1-8B. The manuscript must explicitly confirm that the bootstrap resampling is performed strictly at the problem level for this subset and that the reported standard errors are not inflated by the missing data mechanism. If the number of valid problems varies significantly across candidates, the comparison of means may be biased if not weighted or if the variance estimates do not reflect the differing N .

Second, the choice of the percentile method for confidence intervals assumes the bootstrap distribution is reasonably symmetric. KL divergence and Information Bottleneck residuals are often skewed. The authors should verify the shape of the bootstrap distributions (e.g., via histograms in the appendix) or switch to the Bias-Corrected and Accelerated (BCa) method, which is more robust to skewness and bias. The current symmetric interval formula ($\mu \pm 1.96 \widehat{\sigma}_B$) may be inaccurate if the distribution is non-normal.

Third, the paper presents a large matrix of results (5 models $\times$ ~ 10 candidates $\times$ 4 axioms) and highlights specific instances where a candidate "beats" the baseline (e.g., Table 'tab:per-llm-consistency'). Without a correction for multiple comparisons (such as Bonferroni or Benjamini-Hochberg), the probability of false positives is high. The authors should either apply a correction to the p-values or explicitly state that these findings are exploratory and require validation.

Finally, the correlation analysis between discriminator accuracy and downstream task accuracy ($n=115$) treats each task-model pair as an independent observation. Given that the same model is evaluated across 23 tasks, the residuals are likely correlated within the model. A linear mixed-effects model with a random intercept for the model, or the use of cluster-robust standard errors, would provide a more accurate estimate of the significance of the correlation coefficient.

- [writing] The statistical analysis framework is generally robust, utilizing a large number of bootstrap resamples ($B=10,000$) to estimate uncertainty, which is appropriate for the complex metrics involved (KL divergence, Information Bottleneck residuals). The explicit definition of the statistical unit as the "problem" and the use of cluster bootstrapping for the Causality metric (Section appendix:sub:bootstrap) are strong practices that account for the hierarchical nature of the data (beams within probl

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a high level of technical density, but the writing quality occasionally impedes clarity, particularly in the precise definition of axioms and the flow of complex methodological descriptions.

In Section 2, "Formalizing thought," the definition of the fourth axiom, Stability, is phrased as "T must encode entropy of $P(S|x)$." This sentence is grammatically incomplete and conceptually vague; it is unclear if the representation must **equal** the entropy, **predict** it, or **contain** it. The subsequent sentence, "Quantified by DCS (AUROC of probe predicting $H_x > 0$)," attempts to clarify but does not resolve the ambiguity of the axiom statement itself. The reader is left to infer the relationship between the abstract axiom and the specific metric. A more precise formulation, such as "T must encode the entropy of the semantic distribution $P(S|x)$," would improve readability and scientific precision.

The Conclusion section suffers from a notable sentence fragment: "Measurement cost exceeds single benchmarks." This lacks a clear subject and verb structure, making the statement feel abrupt and unpolished. It should be revised to a complete sentence, for example, "The measurement cost exceeds that of single benchmarks," to maintain the professional tone established in the rest of the paper.

Additionally, in Appendix A.1 ("Bootstrap Confidence Intervals"), the sentence "Beams excluded by the 51-token length gate are absent" is awkwardly phrased. The use of "absent" in this context is non-standard and slightly confusing. A clearer alternative would be "Beams

excluded by the 51-token length gate are omitted from the analysis,” which explicitly states the action taken regarding the data.

Overall, while the paper is readable, these specific instances of ambiguous phrasing and grammatical incompleteness detract from the otherwise rigorous presentation. Addressing these points will enhance the clarity and flow of the manuscript.

- [writing] In Section 2 (Methodology), the definition of Stability (T must encode entropy of $P(S|x)$) is conceptually ambiguous and grammatically disconnected from the subsequent DCS metric description. Clarify the logical link between the axiom statement and the AUROC measurement.
- [writing] The Conclusion section contains a sentence fragment: ‘Measurement cost exceeds single benchmarks.’ This lacks a subject and verb, making the claim unclear. Rephrase to a complete sentence (e.g., ‘The measurement cost exceeds that of single benchmarks.’).
- [writing] In Appendix A.1, the phrase ‘Beams excluded by the 51-token length gate are absent’ is awkward and passive. Suggest rewriting for clarity, such as ‘Beams excluded by the 51-token length gate are omitted from the analysis.’